

Analog Electronic Circuits : Problem Set 8

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For all transistors, use $\mu_n C_{ox} = 200 \mu\text{A}/\text{V}^2$, $\mu_p C_{ox} = 50 \mu\text{A}/\text{V}^2$, $V_{TN} = V_{TP} = 1 \text{ V}$. Use $\lambda = 0$ unless otherwise mentioned. The (W/L)s of the devices are marked next to them.

Problem 1

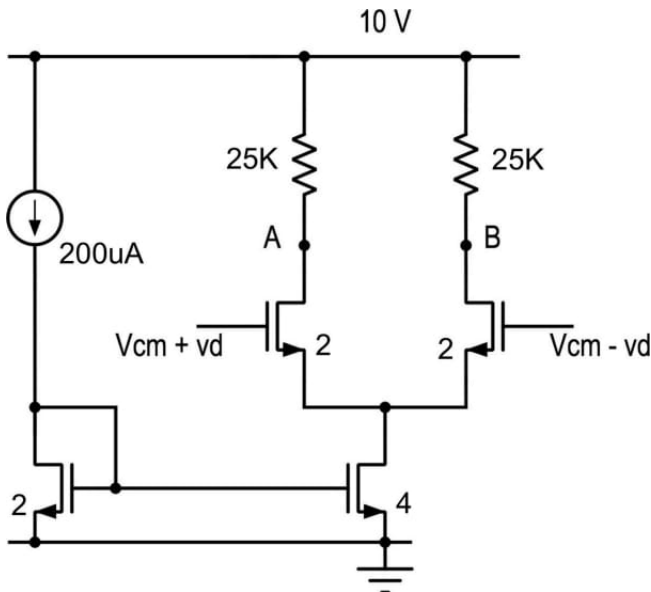


Figure 1: Circuit for problem 1.

In the amplifier of Fig. 1, $V_{cm}=6 \text{ V}$, and v_d is a small signal.

- Determine the operating points of all the devices in the circuit.
- Determine the incremental voltages at A and B.
- What is the range over which V_{cm} can vary, while keeping all devices in saturation?
- It is desired to increase the incremental gain by increasing the resistors, and by using the lowest V_{cm} possible. Determine the maximum achievable incremental gain at node B.

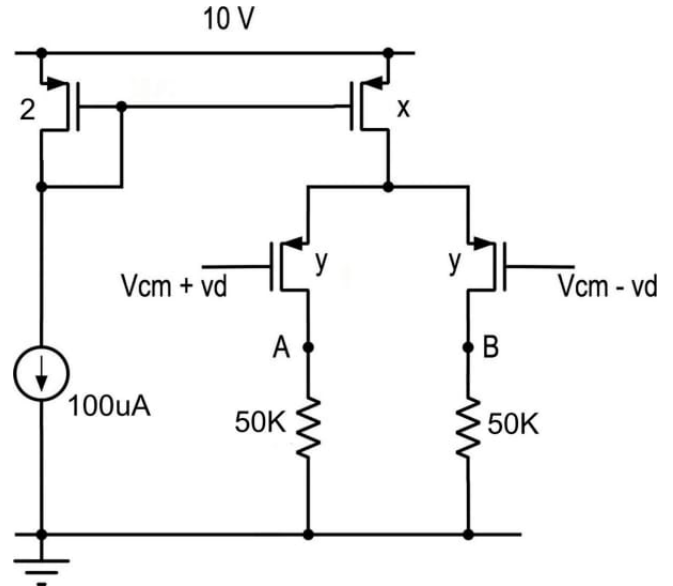


Figure 2: Circuit for problem 2.

Problem 2

In the circuit of Fig. 2, $V_{cm} = 6 \text{ V}$ and v_d is a small signal. Determine the aspect ratios x and y so that

- the quiescent voltages at A and B are the same as in the previous problem, and
- the incremental gain is the same as that in the previous problem.
- With the calculated x and y , find the input common-mode range.

Problem 3

In the circuit of Fig. 3, $V_{cm} = 6 \text{ V}$ and v_d is a small signal. Determine the aspect ratio x , so that the incremental voltage between A and B is $4 v_d$. For this value of x , determine the quiescent voltage at A.

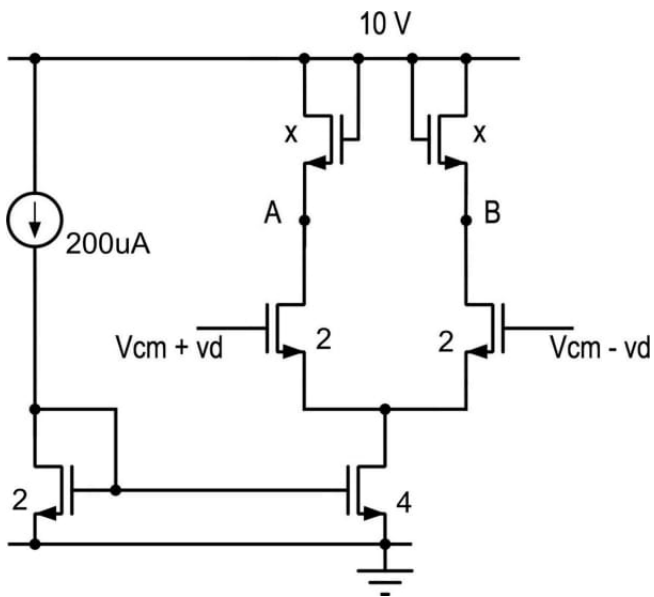


Figure 3: Circuit for problem 3.

Problem 4

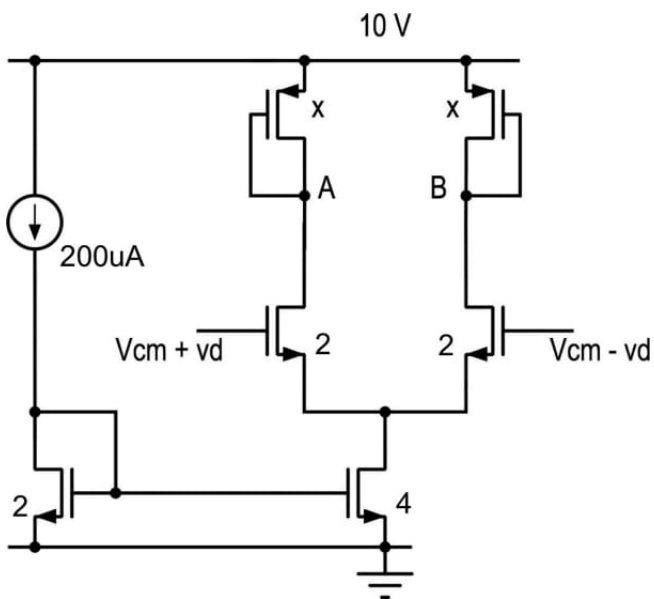


Figure 4: Circuit for problem 4.

Repeat problem 3 for the circuit of Fig. 4.